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Statistical Genomics

Project Report

Statistical Genomics Project

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Abstract

This study investigates the genetic architecture of Attention Deficit Hyperactivity Disorder (ADHD) by integrating genotypic, transcriptomic, and neurochemical data within an endophenotype-based framework. Utilizing a cohort of 642 individuals (275 cases and 367 controls), we analyzed 22 single nucleotide polymorphisms (SNPs) located in key genes of the glutamatergic (GRIN2A, GRIN2B, GRIA1), dopaminergic (DAT, DRD4), and GABAergic signaling pathways. The primary objective was to elucidate the biological pathways bridging genetic variation and clinical behavioral phenotypes.

Our multi-tiered statistical approach included logistic regression for case-control association, quantitative trait locus (QTL) mapping for neurotransmitter levels and gene expression, and pharmacogenomic interaction modeling to assess response to methylphenidate and atomoxetine. Initial predictive modeling using elastic net regression identified potential genetic predictors of ADHD status, while linear models explored cis-regulatory effects of SNPs on gene expression. Furthermore, causal mediation analysis was employed to test the hypothesis that gene expression mediates the relationship between genetic variants and clinical severity (CPRS-R scores). This integrative study highlights the utility of endophenotypes in dissecting the complex etiology of ADHD and provides a preliminary statistical framework for identifying biological markers of disease risk and treatment response.

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1 Introduction

Neurodevelopmental disorders (NDDs) such as Autism Spectrum Disorder (ASD), Attention Deficit Hyperactivity Disorder (ADHD), Intellectual Disability (ID), and Down Syndrome (DS) represent a diverse group of conditions characterised by impairments in cognition, behaviour, communication, and adaptive functioning. These disorders arise from complex interactions between genetic, molecular, and environmental factors, and their etiological pathways often involve dysregulations in neurotransmitter systems, synaptic signaling, and neurodevelopmental gene networks. Understanding these pathways is essential for improving diagnosis, early intervention, and personalized therapeutic decision-making.

Manovikas Kendra (MVK) is a premier multidisciplinary centre in Eastern India dedicated to the assessment, education, therapeutic intervention, and biomedical research related to neurodivergent individuals. Established in 1974, MVK integrates psychiatry, clinical psychology, special education, molecular biology, and pharmacogenomics to offer a holistic framework for understanding neurodevelopmental conditions. The institute operates a Centre for Assessment & Therapy, Centre for Education, Centre for Vocational Training, and a Centre for Molecular & Biomedical Research, the latter recognized as a Scientific and Industrial Research Organisation (SIRO) by the Department of Science & Technology.

Research at MVK has demonstrated that NDDs in the Indian population are strongly influenced by alterations in dopaminergic, serotonergic, glutamatergic, and GABAergic pathways, alongside genetic variants affecting neurotransmitter transporters, metabolizing enzymes, and synaptic receptors. Prior studies from the centre have reported associations between ADHD traits and variations in SLC6A4, GRM7, and CYP2D6, as well as dysregulation of dopaminergic and glutamatergic signaling in both ASD and ADHD probands. These biochemical and genetic markers play a critical role in shaping behavioural traits, treatment responsiveness, and risk stratification.

In this context, the dataset under study provides a rare, integrative genetic–phenotypic snapshot of individuals assessed at MVK, combining genotype information, behavioural trait scores, and pharmacological pre/post intervention data.

1.1 Goal

We try to identify gene markers related to the traits shown by ADHD patients and also perform exploratory analyses on the said markers and traits. MICE has been carried out to perform data imputation for numeric data. The framework and the algorithm used has been explained in detail under the header imputation. Markers which upon imputation have not reached Hardy–Weinberg equilibrium are dropped. An elastic net based model is fitted to the imputed dataset and summary statistics have been visualised for it.

2 Literature Review

Neurodevelopmental disorders such as ASD, ADHD, ID, and DS are characterised by perturbations across glutamatergic, dopaminergic, serotonergic, and GABAergic pathways. The marker panel here spans NMDA/AMPA/kainate subunits (GRIN2A/B, GRIA1, GRIK1), metabotropic glutamate receptors (GRM5/6/7), dopamine transporter/receptor (SLC6A3/DAT, DRD4), and GABA receptor subunits (GABRA5, GABRB3, GABRG3). The rsIDs genotyped in this study (Table 5) map directly onto those loci: rs2229193/rs2284411 (GRIN2A), rs363504/rs363538

(GRIK1), rs1422884/rs2195450 (GRIA1), rs905646/rs11020772/rs762724/rs2067011/rs3792452/rs3749380 (GRM5/6/7), rs1800955 (DRD4), and rs2625511/rs27072 (SLC6A3).

Glutamate pathway (strong evidence). GRIN2A/B (NMDA subunits) and GRIA1 (AMPA) have direct causal roles in excitatory transmission and plasticity; human variants alter receptor kinetics and are linked to neurodevelopmental phenotypes. GRIK1 (kainate) variants such as rs363504/rs363538 show cohort-level associations with ADHD traits and expression differences. GRM5/6/7 (metabotropic) modulate synaptic gain; GRM5/7 variants (e.g., rs905646, rs3792452, rs3749380) are reported in ASD/ADHD cohorts with expression or behavioural associations. Mechanistic support is therefore strong for positive correlations among these genes and with glutamate signalling.

Dopamine pathway (strong evidence). SLC6A3 (DAT; rs2625511, rs27072) governs synaptic dopamine clearance; DRD4 (rs1800955) encodes the D4 receptor with well-described behavioural associations. Both exert direct effects on dopamine tone; network-level modulation of glutamate and GABA is indirect but plausible.

Cross-talk and serotonergic links (moderate/indirect). NMDA and mGluR receptors influence dopaminergic and serotonergic firing patterns in cortico-striatal and limbic circuits, yet direct effects on peripheral 5-HT or 5-HIAA are typically weak. GRM5 interacts pharmacologically with 5-$HT_{2A/2C}$ receptors; glutamatergic variants can therefore be assigned moderate priors for cross-pathway correlations, with wide uncertainty.

GABAergic balance (observational support). Expression of GABRA5, GABRB3, and GABRG3 is elevated in multiple neurodevelopmental cohorts, including the MVK data, consistent with excitation–inhibition compensation. Direct eQTLs from glutamate or dopamine SNPs to these GABA genes are sparse; effects are likely indirect (circuit-level homeostasis).

Implications for modelling. Literature supports strong within-pathway priors: positive correlations among glutamatergic genes and glutamate, and between SLC6A3/DRD4 and dopamine. Cross-pathway priors (glutamate–dopamine, glutamate–5-HT) should be weaker and data-driven. These beliefs motivate the MANOVA, penalised regressions, and classification models, and would map naturally to Bayesian factor or LKJ-based correlation priors if extended.

3 Data Description

3.1 Joint Data Structure and Notation

For each subject $i \in \{1, \ldots, N\}$, define the random vector

$$\mathbf{W}_i := \big(T_i,\ \mathbf{Y}_i,\ \mathbf{G}_i,\ \mathbf{S}_i\big),$$

where

- $T_i \in \{1, \ldots, g\}$ is the *Type* (e.g. ASD, ADHD, ID, DS, control),

- $\mathbf{Y}_i \in \mathbb{R}^p$ is the p-dimensional vector of numeric biomarkers (clinical trait scores, neurotransmitter concentrations, gene-expression levels, etc.),
- $\mathbf{G}_i \in \{0,1,2\}^q$ (or equivalently $\{AA, Aa, aa\}^q$) is the q-dimensional genotype vector (SNP markers),
- $\mathbf{S}_i$ is a vector of other categorical covariates (sex, drug, etc.).

Thus the full dataset can be written as the i.i.d. sample

$$\{\mathbf{W}_i : i = 1, \ldots, N\} \sim P_{T,Y,G,S},$$

a joint distribution on $\{1, \ldots, g\} \times \mathbb{R}^p \times \{0,1,2\}^q \times \mathcal{S}$.

We now separate the different blocks:

$$T := (T_1, \ldots, T_N)^\top, \qquad T_i \in \{1, \ldots, g\},$$
$$Y := (Y_{ij})_{1 \le i \le N,\ 1 \le j \le p} \in \mathbb{R}^{N \times p}, \qquad \mathbf{Y}_i = (Y_{i1}, \ldots, Y_{ip})^\top,$$
$$G := (G_{i\ell})_{1 \le i \le N,\ 1 \le \ell \le q}, \qquad \mathbf{G}_i = (G_{i1}, \ldots, G_{iq})^\top,$$
$$S := (\mathbf{S}_1, \ldots, \mathbf{S}_N)^\top.$$

Table 1: Summary of dataset components and characteristics

Component	**Description**
Subjects	700 total, mixture of Cases and Controls
Genotype markers	Approximately 10–20 SNPs with moderate missingness
Clinical traits	Inattention, Hyperactivity, ADHD Index, Behavioural Problems
Pharmacological data	Pre- and post-treatment trait scores

The methodological pipeline consists of three main steps:

1. Numeric-only multiple imputation of Y via MICE (Section 8.1).
2. SNP-level genotype quality control via Hardy–Weinberg equilibrium (HWE) tests (Section 8.2).
3. One-way MANOVA of $\mathbf{Y}$ on T using Pillai's trace (Section 8.3).

4 Summary Statistics

Descriptive summaries are reported separately for cases and controls. Neurotransmitter concentrations show strong dispersion and marked group differences, particularly for GABA and glutamate. Table 2 provides the key aggregates.

Gene expression summaries (Table 3) indicate higher mean expression of GABA receptor genes in cases, consistent with excitation–inhibition imbalance. Dopaminergic (DRD4) and glutamatergic (GRM5) expressions also diverge between groups.

Table 2: Neurotransmitter concentrations (units as recorded)

oprule Variable	Group	n	Mean	SD	Median	Min–Max
Serotonin	Case	54	31.15	16.31	33.01	1.28–63.85
	Control	26	24.07	9.21	24.28	2.93–44.31
5-HIAA	Case	54	2.07	0.78	1.88	1.10–6.07
	Control	26	2.16	1.03	2.00	1.15–6.03
Dopamine	Case	54	28.08	11.90	23.77	12.25–56.43
	Control	43	32.42	9.10	33.62	13.47–56.04
Glutamate	Case	51	29.34	10.26	29.26	3.76–49.59
	Control	26	36.13	7.04	36.77	16.15–45.86
GABA	Case	54	203.71	195.32	138.73	48.97–1105.50
	Control	26	39.72	20.72	33.66	16.20–121.99

Table 3: Selected gene expression summaries

oprule Gene	Group	n	Mean	SD
GABRB3	Case	42	15.25	2.05
	Control	45	6.67	1.55
GABRA5	Case	42	13.73	1.72
	Control	45	9.17	2.11
GABRG3	Case	43	12.11	4.89
	Control	45	6.38	3.53
GRM5	Case	43	16.62	2.84
	Control	41	11.92	3.44
DRD4	Case	43	9.78	1.22
	Control	43	8.39	1.51
DAT (SLC6A3)	Case	43	14.28	1.96
	Control	41	13.93	2.13

Clinical trait scores are elevated in cases: mean CPRS-R inattention and hyperactivity both exceed 72 (SD $\approx$ 8–13), and ADHD index averages 72.3 (SD 8.5). Post-treatment scores decline by 7–10 points across domains, reflecting measurable pharmacologic benefit.

5 Genomic Analysis

Genotype quality control focused on minor-allele frequency (MAF) and Hardy–Weinberg equilibrium (HWE) within controls. All markers exceeded MAF 0.07, providing sufficient variation. Two loci (M05, M08) failed HWE and were excluded from downstream association models; the remaining markers were retained.

Markers span low- (MAF $\approx$ 0.07) to high-frequency (MAF $\approx$ 0.48) variants; heterozygosity is consequently broad, aiding additive and dominant genetic models. Subsequent analyses (Sections 6 and 10) therefore rely on the filtered panel.

5.1 SNPs and gene mapping

The assayed rsIDs and their gene targets are listed in Table 5. The panel covers ionotropic and metabotropic glutamate receptors, dopamine transport/signalling, and relevant modulators. These rsIDs are used for genotype indicators and drug $\times$ genotype interactions in the downstream elastic-net models.

Table 4: Genotype summary and HWE filter

oprule Marker	MAF	Major/Minor	HWE p	Status
M03	0.07	T/C	0.997	Keep
M06	0.08	C/T	0.289	Keep
M07	0.14	A/G	0.980	Keep
M04	0.16	A/C	0.626	Keep
M01	0.22	C/T	0.998	Keep
M02	0.27	C/T	0.523	Keep
M05	0.34	C/T	5.9×10^{-6}	Drop
M08	0.16	T/G	4.7×10^{-4}	Drop
M22	0.38	C/T	0.968	Keep
M12	0.45	T/C	0.453	Keep
M16	0.47	T/C	0.378	Keep
M09	0.48	T/C	0.081	Keep
M10	0.48	A/G	0.817	Keep
M21	0.48	T/C	0.790	Keep

Table 5: Assayed SNPs and genes

oprule Marker	rsID	Gene
M1	rs2229193	GRIN2A
M2	rs2284411	GRIN2A
M3	rs363504	GRIK1
M4	rs363538	GRIK1
M5	rs1422884	GRIA1
M6	rs2195450	GRIA1
M7	rs905646	GRM5
M8	rs11020772	GRM5
M9	rs762724	GRM6
M10	rs2067011	GRM6
M11	rs3792452	GRM7
M12	rs3749380	GRM7
M16	rs1800955	DRD4
M21	rs2625511	SLC6A3
M22	rs27072	SLC6A3

5.2 Implications

Retaining a broad MAF spectrum preserves power for additive and dominant models while limiting rare-variant sparsity. Excluding HWE-violating M05/M08 reduces spurious associations from potential genotyping error or substructure. The rsID–gene map anchors biological interpretation of elastic-net coefficients and informs pathway-level priors (Section 10).

6 Testing for Effects

Two-sample t-tests compared cases and controls on neurotransmitter concentrations and gene expression. Table 6 lists the strongest effects after Benjamini–Hochberg adjustment; nearly all retain significance at $q < 0.05$.

Findings highlight a dominant excitation–inhibition shift: GABA receptor expression (GABRB3, GABRA5, GABRG3) and GABA concentration are far higher in cases, while glutamate concentration is lower despite

Table 6: Selected case–control t-tests

oprule Variable	Mean (Case)	Mean (Control)	t	BH p
expressions.GABRB3	15.25	6.67	21.90	2.2×10^{-33}
expressions.GABRA5	13.73	9.17	11.10	3.5×10^{-17}
expressions.GRM5	16.62	11.92	6.81	1.0×10^{-8}
expressions.GRM6	12.05	8.62	6.47	2.7×10^{-8}
expressions.GABRG3	12.11	6.38	6.28	6.4×10^{-8}
concentrations.GABA	203.71	39.72	6.10	3.1×10^{-7}
expressions.GRIN2A	13.18	9.93	4.79	2.1×10^{-5}
expressions.DRD4	9.78	8.39	4.67	2.5×10^{-5}
expressions.GRIA1	13.35	10.01	4.54	4.0×10^{-5}
expressions.GRIN2B	12.33	9.16	4.15	1.5×10^{-4}
concentrations.Glutamate	29.34	36.13	−3.41	0.0016
concentrations.Serotonin	31.15	24.07	2.47	0.0204
concentrations.Dopamine	28.08	32.42	−2.03	0.0543

higher GRM5/6/7 and GRIN2A/B expression. Dopamine and serotonin differences are modest and do not survive FDR correction. These univariate contrasts align with MANOVA results and inform the direction of penalised model coefficients discussed later.

7 Basic modeling of Data using OLS

7.1 Gene expressions vs concentration

We try to model gene expressions against concentrations as well as test scores(with and without the drug effect.)

For gene expressions modeled against concentrations, where concentration is treated as a response variable, we come up with the following simple models:

$$T_j = E_{ij}\beta_i + \epsilon_{ij}$$

where

$$T_j\text{: concentration value for j}^{\text{th}}\text{ person}$$
$$E_{ij}\text{: i}^{\text{th}}\text{ gene expression for j}^{\text{th}}\text{ person}$$
$$\epsilon_{ij}\text{: standard error}$$

A little bit more complex model would be the following:

$$T_j = \sum_{i=1}^{E} E_{ij}\beta_i + \epsilon_j$$

There are 5 concentrations, namely 'Serotonin', '5-HIAA','Glutatamate','Dopamine','GABA' and 6 data points.

We shall look at the more complex model, the basic model is given for initial data exploration.

As we shall use LASSO for regularization, we scale the columns to mean 0 and variance 1. Also, since only 6 data points are present, we shall use leave-one-out cross validation technique to tune LASSO.

For serotonin, we have the following plot for mse for different lambda values for LASSO regularization. We choose the local minima for variable selection.

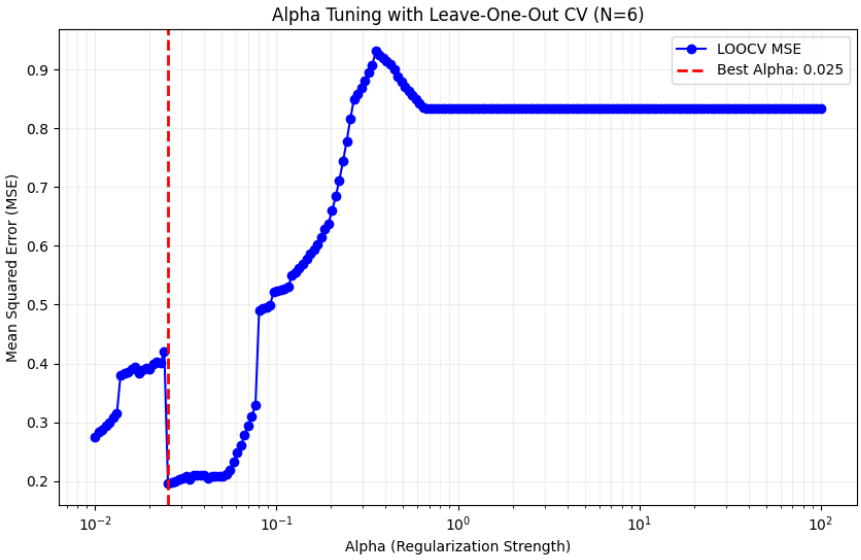


Figure 1: Tuning for LASSO for Serotonin

Coefficients selected by lasso and the values of them are as follows:

Table 7: Coefficients for selected expressions

Expression	Value
expressions.GRIN2A	0.781504
expressions.GRM7	-1.030410
expressions.GRIK1	0.396277
expressions.DAT	-0.126791
expressions.GABRA5	-0.107596

For Dopamine, the following is the resultant alpha and expression coefficients:

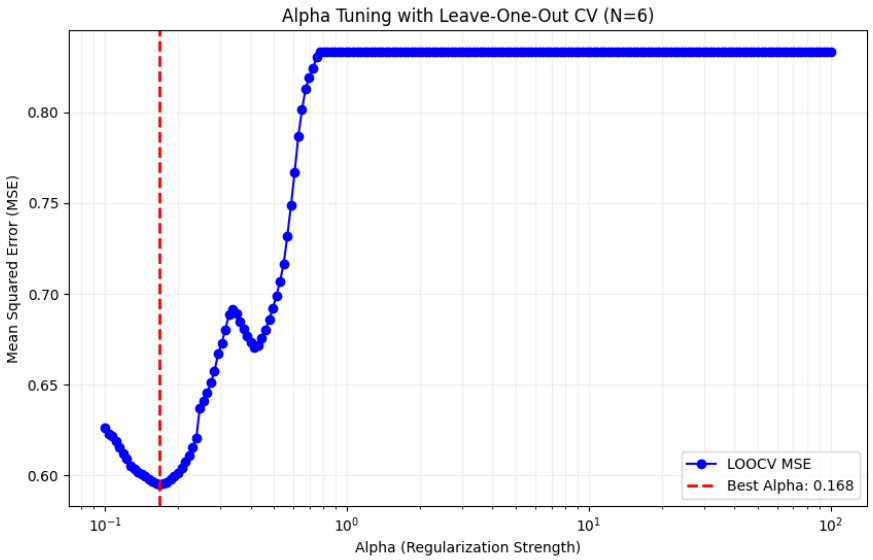


Figure 2: Tuning for LASSO for Dopamine.

Table 8: Coefficients for selected expressions

Expression	Value
expressions.GRIN2A	-0.179528
expressions.DRD4	0.534663

For 5-HIAA, the following is the resultant minimal(local) alpha and the selected coefficients.

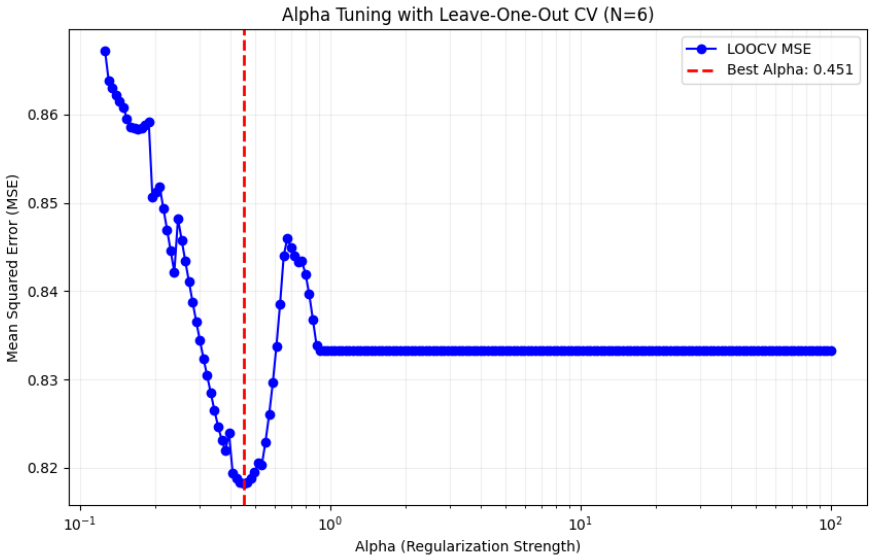


Figure 3: Tuning for LASSO for 5-HIAA.

Table 9: Coefficients for selected expressions

Expression	Value
expressions.GRIN2A	0.291562

For Glutamate, the following is the resultant minimal alpha(local) and the selected coefficients.



Figure 4: Tuning for LASSO for Glutamate.

Table 10: Coefficients for selected expressions

Expression	Value
expressions.GRM5	-0.377961

LASSO fails to regularize against GABA as MSE decreases with decreasing alpha/lambda without having any local minima. More data points are required to successfully draw inference for concentration of GABA against various expressions.

7.2 Gene expressions vs test scores(with drug effect)

We seek to infer the effect of gene expressions on the effectivity of the drug. For this, we first write the models:

$Y_{i0} = \mu + \sum_{i=1}^{E} E_{ij}\beta_j + \epsilon_i 0$ $Y_{i1} = \mu + \sum_{i=1}^{E} E_{ij}\beta_j^* + Z_i + \epsilon_i 1$

Now, we only have 5 datapoints for methylphenidate as drug and two datapoints for atomoxetin.Since we cannot draw inference based on two datapoints, we only take drug methylphenidate into consideration. This causes the mixed model term Z_i to become $\mu^* + \epsilon$ and the epsilon gets added to the error term.

Subtracting the two equations, we get out linear model: $\Delta Y = Y_{i1} - Y_{i0} = \mu^* + \sum_{i=1}^{E} E_{ij}(\beta_j^* - \beta_j) + \epsilon'$ which can be rewritten as $\Delta Y = \mu^* + \sum_{i=1}^{E} E_{ij}(\beta_j') + \epsilon'$.

The test scores are respectively for 'Behavorial Problem','Inattention','Hyperactivity' and 'ADHD Index'.

Since we have only 5 data points, we shall use lasso with leave one out method of cross-validation for fine-tuning.

For Behavorial problem,the following is the resultant fine tuned parameter and the ones after it are the selected coefficients.

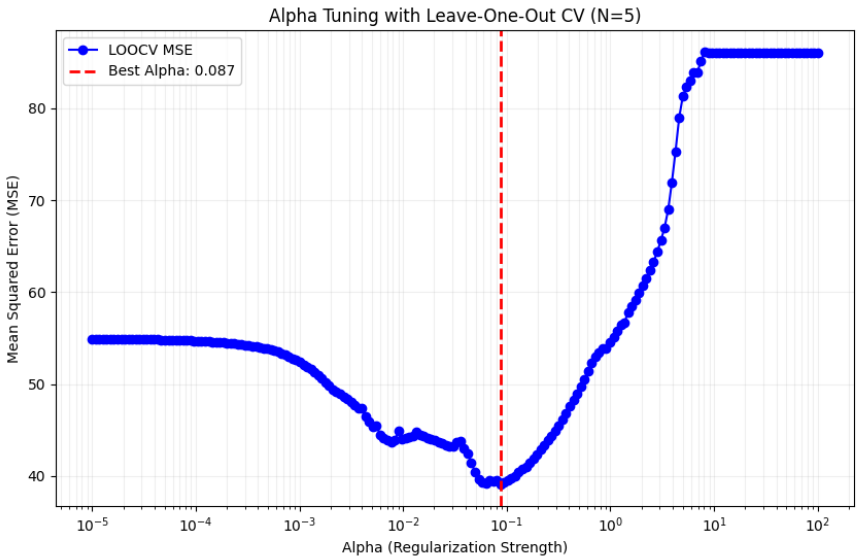


Figure 5: Tuning for LASSO for Behavorial Problems

Table 11: Selected values

Variable	Coefficient
μ^*	-7.112964
expressions.GRIA1	1.080089
expressions.GRIN2B	1.771589
expressions.GRM5	0.130357
expressions.GRIK1	2.881464
expressions.GABRG3	-1.794950

For Hyperactivity, the following are the resultant fine tuned parameter and the selected coefficients:

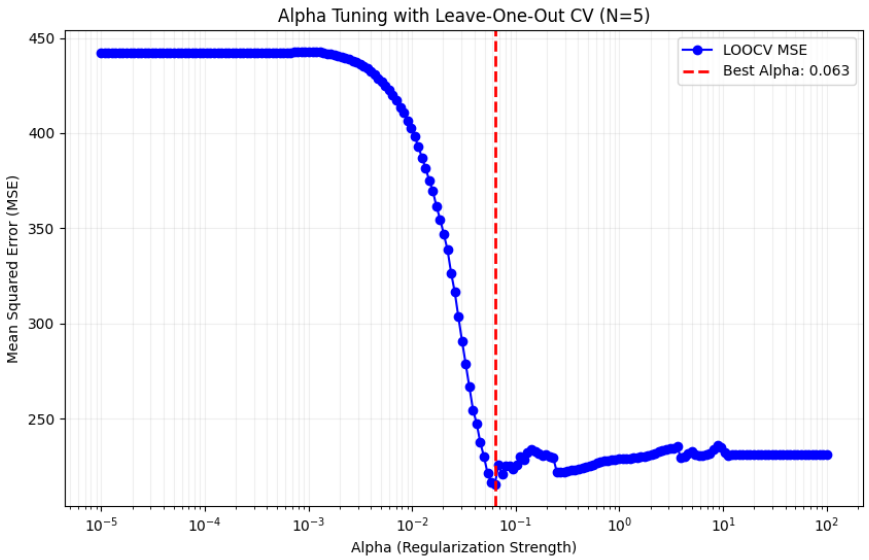


Figure 6: Tuning for LASSO for Hyperactivity

Table 12: Selected Values

Term	Coefficient
μ^*	-10.137050
expressions.GRIA1	5.388715
expressions.GRM6	-3.464527
expressions.GRIK1	2.945952
expressions.DAT	-5.002425
expressions.GABRB3	-6.391410
expressions.GABRG3	-3.745889

For ADHD index, we have the following results:

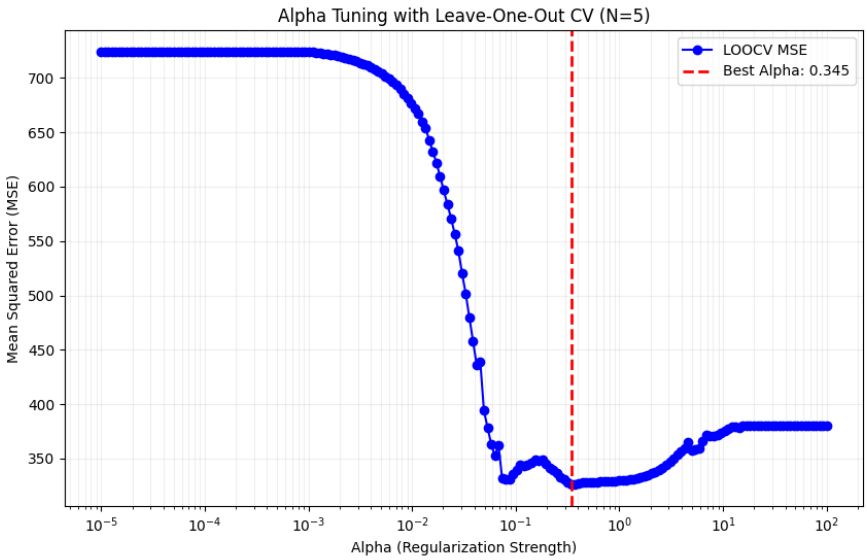


Figure 7: Tuning for LASSO for ADHD index

Table 13: Selected values

Variable	Coefficient
μ^*	-13.055104
expressions.GRIA1	7.511725
expressions.DAT	-0.174748
expressions.GABRB3	-14.256377
expressions.GABRG3	-3.937476

For inattention,unlike the other test scores we don't get a global minima but have to take a local minima as the optimal value for alpha.Upon doing so we get all the coefficients removed apart from μ^* which is the constant term. Thus, LASSO fails to regularize inattention and more data points are required for inference.

7.3 Gene expressions vs Test scores (Without drug effect)

This scenario has some benefit to study in the sense that there are 20 data points which is a great improvement over single digit data strength. We deal with this model only now:

$$Y_{i0} = \mu + \sum_{i=1}^{E} E_{ij}\beta_j + \epsilon_i 0$$

We shall be using elastic net regression with 5 fold cross-validation for hyperparameter tuning (both alpha and L1-weight).

Cprs-r.Behavioral Problem: Best Alpha: 0.2364, Best L1 weight: 1.0000, Min MSE: 94.9563

Table 14: Selected Variables

Variable	Value
μ	67.413551
expressions.GRIN2A	-11.216847
expressions.GRM5	-0.292382
expressions.GRM7	3.510165
expressions.GRIK1	4.071858
expressions.DRD4	-0.741441
expressions.GABRB3	1.477106
expressions.GABRG3	9.193305
expressions.GABRA5	-1.114006

Cprs-r.Inattention: Best Alpha: 3.5112, Best L1 weight: 1.0000, Min MSE: 328.4904

Table 15: Selected Variables

Variable	Value
μ	68.938808
expressions.GRM5	1.583019

Cprs-r.Hyperactivity: Best Alpha: 0.0231, Best L1 weight: 0.4898, Min MSE: 109.1093

Table 16: Selected Variables

Variable	Value
μ	69.815810
expressions.GRIA1	13.730005
expressions.GRIN2A	-15.278839
expressions.GRIN2B	-7.579745
expressions.GRM5	9.371283
expressions.GRM6	-7.292428
expressions.GRIK1	0.011715
expressions.DAT	2.556493
expressions.DRD4	0.973209
expressions.GABRB3	-2.539126
expressions.GABRG3	8.502564
expressions.GABRA5	0.162945

Cprs-r.ADHD Index: Best Alpha: 1.5999, Best L1 weight: 1.0000, Min MSE: 113.3826

Table 17: Selected Variables

Variable	Value
μ	72.650141
expressions.GRM5	2.150134
expressions.GRIK1	0.113296

8 Multiple Imputation and Multivariate Analysis

8.1 Step 1: Multiple Imputation by Chained Equations (MICE)

We have done numeric-only multiple imputation (MICE) on the dataset, while excluding genotype, sex, drug, type, and ID variables from being used as predictors or targets. The goal is to fill in missing numeric data responsibly without bias from categorical or identifier variables.

8.1.1 Missingness Structure and MAR Assumption

Let $R = (R_{ij})$ be the missingness indicator for the numeric block Y:

$$R_{ij} = \begin{cases} 1, & \text{if } Y_{ij} \text{ is observed,} \\ 0, & \text{if } Y_{ij} \text{ is missing.} \end{cases}$$

For each $j \in \{1, \ldots, p\}$, define the sets of observed and missing indices:

$$\mathcal{O}_j := \{i \in \{1, \ldots, N\} : R_{ij} = 1\}, \qquad \mathcal{M}_j := \{i \in \{1, \ldots, N\} : R_{ij} = 0\}.$$

We assume a *Missing At Random* (MAR) mechanism:

$$P(R \mid Y, G, T, S) = P\big(R \mid Y_{\text{obs}}, G, T, S\big),$$

and in the actual imputation models we condition only on Y; the blocks G, T, S are not used as predictors nor imputed.

8.1.2 MICE Algorithm Framework

For this purpose, we have used the following MICE framework:

8.1.3 Predictive Mean Matching (PMM)

For the final step we have used the following PMM method:

Algorithm 1 MICE algorithm for imputation of multivariate missing data

Specify an imputation model $P(Y^j_{mis} \mid Y^j_{obs}, Y_{-j}, R)$ for variable Y_j with $j = 1, \ldots, p$.
for each j **do**
    Fill in starting imputations $\dot{Y}^0_j$ by random draws from Y^j_{obs}.
end for
for $t = 1, \ldots, M$ **do**
    for $j = 1, \ldots, p$ **do**
        Define $\dot{Y}^t_{-j} = (\dot{Y}^t_1, \ldots, \dot{Y}^t_{j-1}, \dot{Y}^{t-1}_{j+1}, \ldots, \dot{Y}^{t-1}_p)$ as the currently complete data except Y_j.
        Draw $\dot{\phi}^t_j \sim P(\phi^t_j \mid Y^j_{obs}, \dot{Y}^t_{-j}, R)$.
        Draw imputations $\dot{Y}^t_j \sim P(Y^j_{mis} \mid Y^j_{obs}, \dot{Y}^t_{-j}, R, \dot{\phi}^t_j)$.
    end for
end for

Algorithm 2 PMM with a Bayesian β and a stochastic matching distance

Calculate the cross-product matrix $S = X'_{obs}X_{obs}$.
Calculate $V = (S + \text{diag}(S)\kappa)^{-1}$, with some small κ.
Calculate regression weights $\widehat{\beta} = VX'_{obs}Y_{obs}$.
Draw a random variable $\dot{g} \sim \chi^2_\nu$ with $\nu = n_1 - p + 1$.
Calculate $\dot{\sigma}^2 = \frac{(Y_{obs} - X_{obs}\widehat{\beta})'(Y_{obs} - X_{obs}\widehat{\beta})}{\dot{g}}$.
Draw $p - 1$ independent $N(0, 1)$ variates in vector $\dot{z}_1$.
Calculate $V^{1/2}$ by Cholesky decomposition.
Calculate $\dot{\beta} = \widehat{\beta} + \dot{\sigma}\dot{z}_1V^{1/2}$.
Calculate $\dot{\eta}(i, j) = |(X_{obs})_{i\cdot}\widehat{\beta} - (X_{mis})_{j\cdot}\dot{\beta}|$ with $i = 1, \ldots, n_1$ and $j = 1, \ldots, n_0$.
Construct n_0 sets Z_j, each containing d candidate donors (here, $d = 5$), from Y_{obs} such that $\sum_d \dot{\eta}(i, j)$ is minimum for all $j = 1, \ldots, n_0$. Break ties randomly.
Draw one donor i_j from Z_j randomly for $j = 1, \ldots, n_0$.
Calculate imputations $\dot{Y}_j = Y_{i_j}$ for $j = 1, \ldots, n_0$.

This imputation procedure will be useful when we try to see the gene-drug interaction via an elastic net procedure.

8.2 Step 2: Genotype Markers and HWE Filtering

We now focus on the genotype block $G = (G_{i\ell})$ and its relation to Type T. This step is a quality-control filter on the SNP markers, independent of the MICE and MANOVA steps.

Let $\mathcal{C} \subset \{1, \ldots, N\}$ denote the index set of control subjects (e.g. T_i corresponding to "control"). For marker $\ell \in \{1, \ldots, q\}$:

- Observed genotypes in controls:
$$\{G_{i\ell} : i \in \mathcal{C}\} \subset \{AA, Aa, aa\}.$$
- Genotype counts:
$$n_{g,\ell} := \sum_{i \in \mathcal{C}} \mathbf{1}\{G_{i\ell} = g\}, \quad g \in \{AA, Aa, aa\},$$
and total count $n_\ell := \sum_g n_{g,\ell}$.

- Minor-allele frequency (for allele a):

$$p_\ell := \frac{2n_{aa,\ell} + n_{Aa,\ell}}{2n_\ell}.$$

Under the Hardy–Weinberg equilibrium (HWE) null hypothesis

$$H_{0,\ell}: \quad P(AA) = (1-p_\ell)^2, \quad P(Aa) = 2p_\ell(1-p_\ell), \quad P(aa) = p_\ell^2,$$

the expected genotype counts are

$$E_{AA,\ell} = n_\ell(1-p_\ell)^2, \quad E_{Aa,\ell} = 2n_\ell p_\ell(1-p_\ell), \quad E_{aa,\ell} = n_\ell p_\ell^2.$$

Define the Pearson chi-square test statistic

$$X_\ell^2 := \sum_{g\in\{AA,Aa,aa\}} \frac{\left(n_{g,\ell} - E_{g,\ell}\right)^2}{E_{g,\ell}} \overset{H_{0,\ell}}{\approx} \chi_1^2.$$

Markers with $p_\ell^{(\mathrm{HWE})} < \alpha_{\mathrm{HWE}}$ (e.g. $\alpha_{\mathrm{HWE}} = 10^{-4}$) are removed from further genetic analyses.

8.3 Step 3: One-Way MANOVA for Type Effect

We now consider a fixed imputed numeric dataset Y (one of the $Y^{(m)}$), together with the group labels $T = (T_1, \ldots, T_N)^\top$. We perform a one-way MANOVA to test whether the multivariate mean biomarker profiles differ across Types.

8.3.1 Model

For $i = 1, \ldots, N$, we model

$$\mathbf{Y}_i = \boldsymbol{\mu}_{T_i} + \boldsymbol{\varepsilon}_i, \qquad \boldsymbol{\varepsilon}_i \sim \mathcal{N}_p(\mathbf{0}, \Sigma),$$

where

- $\boldsymbol{\mu}_k \in \mathbb{R}^p$ is the mean biomarker vector for Type k,
- $\Sigma \in \mathbb{R}^{p\times p}$ is a common covariance matrix across all Types,
- $\boldsymbol{\varepsilon}_i$ are i.i.d. across i.

The hypothesis of interest is

$$H_0: \ \boldsymbol{\mu}_1 = \cdots = \boldsymbol{\mu}_g \quad \text{vs.} \quad H_A: \ \exists\, k \neq k' \text{ with } \boldsymbol{\mu}_k \neq \boldsymbol{\mu}_{k'}.$$

8.3.2 Pillai–Bartlett Trace and Approximate F-Test

The Pillai–Bartlett trace is one of the four classical test statistics used in multivariate analysis of variance (MANOVA). It evaluates the combined effect of the independent variables on multiple dependent variables simultaneously.

Let $\mathbf{H}$ denote the hypothesis sum of squares and cross-products (SSCP) matrix, and $\mathbf{E}$ the error SSCP matrix. The eigenvalues λ_i are obtained from the characteristic equation of $\mathbf{E}^{-1}\mathbf{H}$. The Pillai's trace statistic V is then defined as

$$V = \text{trace}\big[\,\mathbf{H}(\mathbf{H}+\mathbf{E})^{-1}\big] = \sum_{i=1}^{s} \frac{\lambda_i}{1+\lambda_i},$$

where $s = \min(p, g-1)$, p is the number of dependent variables, and g the number of groups.

Under the null hypothesis $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2 = \cdots = \boldsymbol{\mu}_g$, the statistic V is transformed into an approximate F statistic:

$$F = \frac{(2n+p+1)V}{(2m+2-2V)},$$

with corresponding numerator and denominator degrees of freedom depending on p, g, and sample size N.

A larger value of V (and consequently of F) provides evidence against the null hypothesis, indicating that the group mean vectors differ significantly in the multivariate space.

8.4 Empirical Results from the MVK Dataset

Applying the above pipeline to the MVK dataset:

1. Numeric variables (clinical traits, neurotransmitter levels, gene-expression levels) were imputed using numeric-only MICE, yielding fully completed matrices $Y^{(m)}$.
2. Genotype markers were subjected to HWE filtering; SNPs with extreme deviations from HWE were removed from subsequent genetic analyses.
3. On a representative imputed dataset Y, a one-way MANOVA was performed to test the global effect of Type on the joint biomarker profile, using Pillai's trace.

The resulting multivariate test was:

$$V \approx 0.9681, \qquad F(17, 682) \approx 1217.3, \qquad p < 2.2 \times 10^{-16}.$$

This extremely large Pillai trace and F-statistic, with essentially zero p-value, provide overwhelming evidence against H_0, indicating that the multivariate mean vectors $\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_g$ associated with the different Types differ substantially in the p-dimensional biomarker space.

Univariate decompositions of H and E (Table 18) suggest that biomarkers such as GABA concentrations, GABRB3 expression, glutamate, serotonin, and GRM5 expression contribute most strongly to the group separation, while some other variables (e.g. 5-HIAA, DAT, GRIK1) contribute less.

9 Case-Control Discrimination

9.1 Data Preprocessing and Imputation

Prior to statistical modeling, both the Concentration and Gene Expression datasets underwent rigorous preprocessing to address missing data and standardize the structure for classification.

Table 18: MANOVA Results: Analysis of Variance for Type

Response Variable	Sum of Squares		Residual Std. Err.
	Type ($df = 1$)	Residuals ($df = 698$)	
Gene Expressions			
GRIA1	1726	1599	1.513
GRIN2A	1606	1424	1.428
GRIN2B	1685	1883	1.642
GRM5	3978	1576	1.503
GRM6	1748	1046	1.224
GRM7	1248	1836	1.622
GRIK1	90	608	0.934
DAT	17	930	1.154
DRD4	329	374	0.732
GABRB3	12,889	631	0.951
GABRG3	6321	2865	2.026
GABRA5	3665	723	1.018
Concentrations			
Serotonin	4829	36,668	7.248
5-HIAA	5	133	0.437
Glutamate	7646	14,971	4.631
Dopamine	3286	23,922	5.854
GABA	3,520,778	2,772,917	63.029

Table 19: MANOVA Test Results (Pillai's Trace)

Source	Df	Pillai	Approx F	Num Df	Den Df	Pr(>F)
Type	1	0.9681	1217.3	17	682	$< 2.2 \times 10^{-16}$ ***
Residuals	698					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

9.1.1 Concentrations Dataset

- **Row Deletion:** Rows containing missing values for 4 or more compounds (out of 5 possible) were removed to ensure data quality.
- **Imputation:** Remaining missing values were imputed using the column mean of the available data.
- **Transformation:** The identifier variable `ID` was transformed into a binary response variable: **0** for Control (IDs starting with 'C') and **1** for all other IDs.

9.1.2 Gene Expressions Dataset

- **Row Deletion:** Rows with 6 or more missing gene values were excluded.
- **Imputation:** For the remaining samples, missing values were imputed using the column means computed from the filtered dataset.
- **Transformation:** The same binary transformation was applied to the `ID` variable (0 = Control, 1 = Others).

9.2 Logistic Regression and Complete Separation

Ordinary Logistic Regression failed to converge for both datasets. The models yielded infinite coefficients and standard errors, accompanied by singularity warnings. This failure was caused by the phenomenon of **Complete Separation**. In both datasets, the predictors were highly effective at distinguishing the two groups. Mathematically, the data points for Group 0 and Group 1 could be perfectly separated by a linear combination of the predictors (a hyperplane).

- **Concentration Data:** The compound **GABA** alone provided a near-perfect separation threshold.
- **Expression Data:** The core genes (**GRM5, GRIN2A**) provided a distinct boundary.

Because the groups were perfectly distinct, the Maximum Likelihood Estimation (MLE) algorithm attempted to drive the predicted probabilities to exactly 0 and 1. To achieve this, the regression coefficients grew indefinitely towards infinity, causing the optimization algorithm to crash.

Table 20: Example of Failed Logistic Model Summary (Concentration Data)

Variable	Coefficient	Std. Error	z-score	P-value
Intercept	0.8019	2411.865	0.000	1.000
Serotonin	0.9157	38.436	0.024	0.981
5-HIAA	-5.2325	1324.943	-0.004	0.997
Glutamate	-0.0022	0.078	-0.028	0.978
Dopamine	-1.1533	48.744	-0.024	0.981
GABA	**1537.78**	**29,500.00**	**0.052**	**0.958**

9.3 Linear Discriminant Analysis (LDA) Results

To overcome the convergence issues of logistic regression, Linear Discriminant Analysis (LDA) was employed to find the linear combination of features that best separates the two classes.

9.3.1 Concentration Data

The coefficient for GABA was extremely large and positive (+8.51). This indicates that higher GABA concentrations are the primary driver classifying a sample as **Group 1** (Non-Control). Other compounds played negligible roles by comparison.

Table 21: LDA Coefficients (Concentration Data)

Feature	Coefficient	Interpretation
GABA	**8.5119**	**Strong Positive Predictor (Group 1)**
Serotonin	0.0239	Weak Positive
5-HIAA	-0.3596	Weak Negative
Dopamine	-0.0911	Weak Negative
Glutamate	-0.0001	Negligible

9.3.2 Gene Expression Data

Analysis revealed that higher expression levels of *GRIN2A* and *GRM5* are strongly associated with **Group 1**. Conversely, *GRM6* and *DRD4* had strong negative coefficients, suggesting higher levels of these genes are associated with **Group 0** (Control).

Table 22: LDA Coefficients (Gene Expression Data)

Gene	Coefficient	Interpretation
GRIN2A	**1.5597**	**Strong Positive Predictor (Group 1)**
GRM5	**1.4336**	**Strong Positive Predictor (Group 1)**
GABRA5	0.7214	Moderate Positive
GABRG3	0.2348	Weak Positive
GRM7	0.2172	Weak Positive
GRIN2B	-0.1032	Weak Negative
DAT	-0.1638	Weak Negative
GRIA1	-0.2396	Moderate Negative
GRIK1	-0.2583	Moderate Negative
GABRB3	-0.2609	Moderate Negative
DRD4	**-0.8965**	**Strong Negative Predictor (Group 0)**
GRM6	**-1.2334**	**Strong Negative Predictor (Group 0)**

9.4 Tests of Matrix Equality

To validate the statistical assumptions of our classification models, we tested for the equality of variance-covariance and correlation matrices between the two groups.

Table 23: Tests of Matrix Equality Results

Dataset	Test Type	Metric	P-Value	Interpretation
Concentration	Covariance	Box's M	< 0.001	**Reject** H_0: Variances differ
Concentration	Correlation	Permutation	0.275	**Fail to Reject** H_0: Correlations stable
Expression	Covariance	Box's M	< 0.001	**Reject** H_0: Variances differ
Expression	Correlation	Permutation	0.137	**Fail to Reject** H_0: Correlations stable

The results consistently show that while the **variances** differ significantly between groups (magnitude of signal), the **correlations** (structure of relationships) remain statistically stable.

9.5 Visualizing Separation

Visual inspection of the most prominent variables identified by LDA confirmed the strong separation found in the models. The "Best Separation Value" indicates the numerical threshold that minimizes classification error.

Table 24: Separation Analysis Summary

Dataset	Variable	Best Sep. Value	Interpretation
Concentration	GABA	0.0532 ng/ml	Samples > 0.0532 are almost exclusively Group 1.
Expression	GRM5	10.07	Controls show expression levels < 10.07.
Expression	GRIN2A	8.37	Controls show expression levels < 8.37.

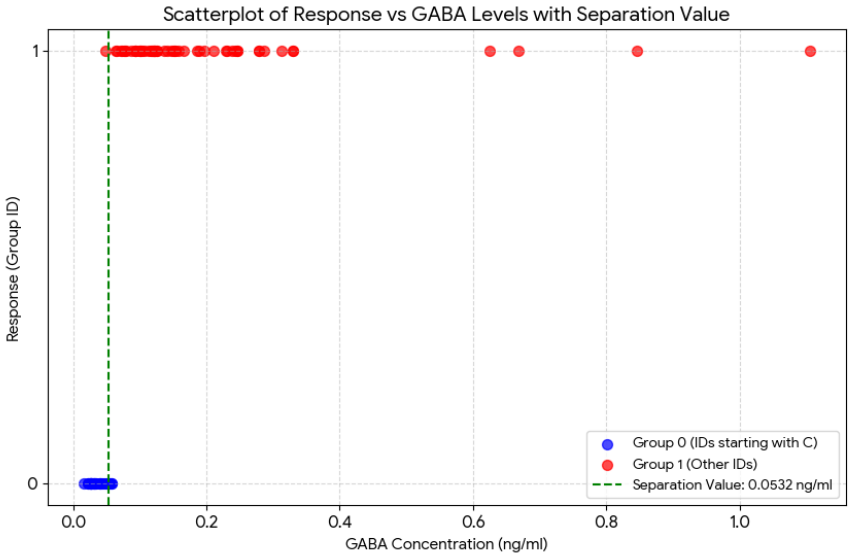


Figure 8: Scatterplot of Response vs. GABA Concentration. The green dashed line represents the optimal separation threshold.

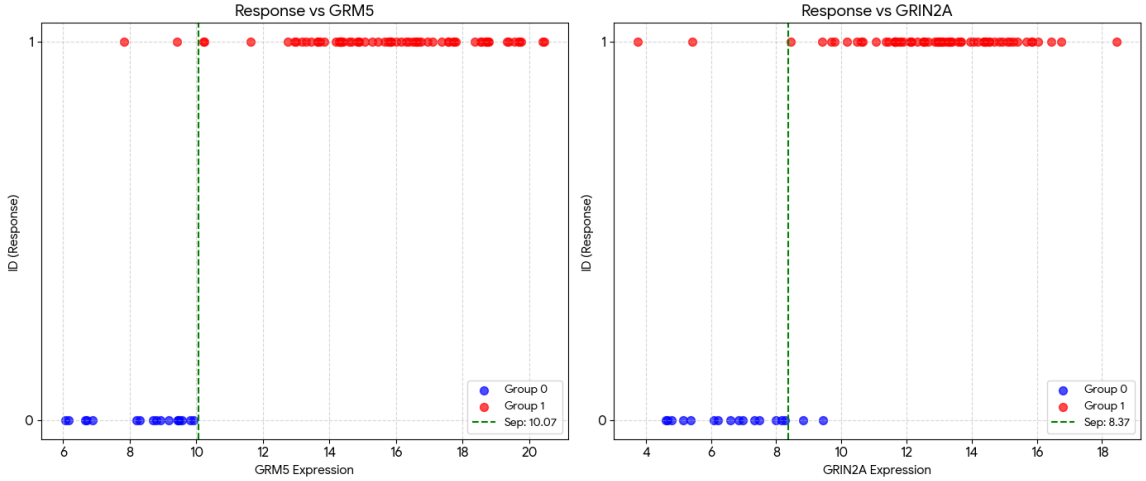


Figure 9: Scatterplots of Response vs. GRM5 and GRIN2A Expression. These genes individually provide excellent discrimination between the groups.

9.5.1 2D LDA Decision Boundary

Using the two most critical genes, GRM5 and GRIN2A, we visualized the decision boundary created by the Linear Discriminant Analysis model.

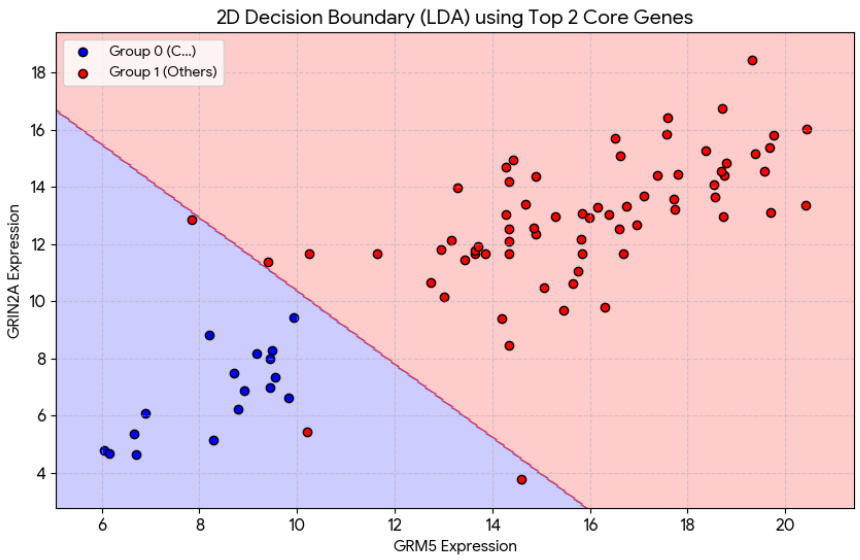


Figure 10: 2D Decision Boundary (LDA) using Top 2 Core Genes (GRM5 and GRIN2A).

9.6 LASSO Penalized Logistic Regression

To address the stability issues of ordinary logistic regression, a LASSO (L1-penalized) model was fitted to the Gene Expression data. The regularization path highlighted the most robust genetic predictors as the penalty parameter (λ) was varied.

Table 25: LASSO Variable Selection Path (Gene Expression)

Penalty $(1/\lambda)$	**No. of Genes**	**Selected Genes**
High (≈ 0.04)	1	**GRM5**
Moderate (≈ 0.07)	2	**GRIN2A, GRM5**
Low (≈ 0.11)	3	GRIA1, GRIN2A, GRM5
Very Low (< 0.01)	10+	All genes included (DAT, DRD4, etc.)

Conclusion: The analysis indicates that the two groups are biologically distinct with high statistical certainty. The separation is driven primarily by **GABA** levels in the concentration data and by **GRM5/GRIN2A** overexpression in the genetic data. The failure of ordinary logistic regression was a direct consequence of this extremely strong biological signal.

10 Elastic Net Regression for Post-Treatment Outcome

On each imputed dataset we fit an elastic-net linear model for post-treatment ADHD index, with baseline scores, neurotransmitter concentrations, expression, CPRS-R subscales, drug indicators, genotype, and drug × genotype interactions as predictors. Columns were standardized; only the intercept was unpenalized. Tuning used K-fold CV over (α, λ). Representative coefficients at the chosen λ are shown in Tables 26 and 27.

10.1 Model

For each completed imputed dataset $D^{(m)}$ (numeric covariates imputed; categorical variables and the outcome not imputed), we fit an elastic–net penalized linear model for the post–treatment ADHD score $Y_i =$ `pharma.post.ADHD Index`. For subject $i = 1, \ldots, n$, define the predictor vector

$$x_i = \big[1,\ B_i,\ E_i,\ C_i,\ R_i,\ D_i,\ G_i,\ (D_i \otimes G_i)\big],$$

where $B_i =$ `pharma.pre.ADHD Index` (baseline severity), E_i stacks the expression features `expressions.*`, C_i stacks the neurotransmitter concentrations `concentrations.*`, R_i stacks the CPRS–R scores `cprs-r.*`, D_i are one–hot drug indicators, G_i are one–hot genotype indicators (reference levels omitted), and $D_i \otimes G_i$ are all drug×genotype interaction indicators. Let $X \in \mathbb{R}^{n \times p}$ collect these predictors without the intercept; all numeric columns are standardized within $D^{(m)}$. The elastic–net estimator is

$$(\widehat{\beta}_0^{(m)}, \widehat{\beta}^{(m)}) \in \arg\min_{\beta_0, \beta}\ \frac{1}{n}\big\|Y - \beta_0 \mathbf{1} - X\beta\big\|_2^2\ +\ \lambda\left[(1-\alpha)\frac{\|\beta\|_2^2}{2} + \alpha\|\beta\|_1\right],$$

with mixing parameter $\alpha \in [0, 1]$ and penalty level $\lambda > 0$; only the intercept is unpenalized. Tuning uses K-fold cross–validation to select $\lambda_{\min}$ or $\lambda_{1\mathrm{SE}}$ (optionally over a grid of α). Predictions on $D^{(m)}$ are

$$\widehat{Y}_i^{(m)} = \widehat{\beta}_0^{(m)} + x_i^\top \widehat{\beta}^{(m)}.$$

When multiple imputations are used ($m = 1, \ldots, M$), this fit is repeated per m and results are summarized across imputations. Here we run the imputation only once so $m = 1$.

10.2 Elastic–Net Outputs (Impute–Then–Analyze)

- *Selected features (support):* the set of predictors with nonzero coefficients at the tuned penalty, including key Drug×Genotype interactions.
- *Coefficient estimates:* signs and magnitudes $(\widehat{\beta})$ for selected terms; when multiple imputations are used, summarize by median and IQR across completed datasets.
- *Tuned hyperparameters:* the chosen mixing $\alpha \in [0, 1]$ and penalty level $\lambda > 0$ (typically $\lambda_{\min}$ or $\lambda_{1\mathrm{SE}}$ from K-fold cross–validation).
- *Generalization metrics:* cross–validated RMSE and R^2, reported per imputation and aggregated (mean ± SD).
- *Predictions & residuals:* fitted values $\widehat{Y}_i = \widehat{\beta}_0 + x_i^\top \widehat{\beta}$ and residuals $e_i = Y_i - \widehat{Y}_i$ for calibration/error analysis.

- *Stability summaries:* selection frequencies of each predictor across imputations (and across α), identifying a stable core set of features.
- *Post–selection inference (optional):* refit on the stable support to report coefficient estimates, confidence intervals, and p-values for interpretability.
- *Diagnostics/visuals:* coefficient paths vs $-\log\lambda$, CV error curves vs $\log\lambda$, performance vs α, observed–vs–predicted plots, and residual distributions.

10.3 CV GLMNET Results

10.3.1 Neurotransmitter Concentrations

Table 26: Elastic net coefficients for neurotransmitter concentrations

Variable	Coefficient
concentrations.5-HIAA	-0.1098
concentrations.Dopamine	-0.0866
concentrations.Glutatamate	-0.0774
concentrations.GABA	0.0494
concentrations.Serotonin	0.0304

Three neurotransmitter concentrations (5-HIAA, dopamine, glutamate) are negatively related to the dependent variable. Two (GABA and serotonin) show a positive relationship. The magnitude of the coefficients suggests the effect sizes are relatively small, with 5-HIAA being the largest in absolute value.

Minimum mean CV error: ≈ 0.547 at $\lambda \approx 0.001577$, with all 5 concentration variables selected (nzero = 5).

10.3.2 Gene Expressions

Table 27: Elastic net coefficients for gene expressions

Variable	Coefficient
expressions.GABRB3	1.1250
expressions.GABRG3	0.4237
expressions.GRIK1	-0.3921
expressions.GABRA5	0.3220
expressions.DAT	0.1171
expressions.GRM7	-0.0305
expressions.GRIA1	0
expressions.GRIN2A	0
expressions.GRIN2B	0
expressions.GRM5	0
expressions.GRM6	0
expressions.DRD4	0

Expression of GABA-related genes (GABRB3, GABRG3, GABRA5) strongly predicts the outcome. GRIK1 (glutamate receptor gene) is the main negative driver. Many genes were shrunk to zero, meaning strong collinearity or weak signals relative to selected genes.

The model favors GABAergic expression (positive) and penalizes glutamatergic GRIK1. Dopamine and serotonin concentrations have small positive effects; 5-HIAA, dopamine, and glutamate are negative.

Minimum CV error ≈ 0.1155 occurs around $\lambda \approx 0.00235$, with 6 genes selected (nzero = 6).

11 Some Other Models for Traits

11.1 Concentrations vs. expression: sparse linear models

Leave-one-out LASSO models related neurotransmitter concentrations to expression features. Serotonin was driven by GRIN2A, GRM7, GRIK1 (positive), with negative DAT and GABRA5 effects. Dopamine depended on GRIN2A (negative) and DRD4 (positive). 5-HIAA selected GRIN2A alone; glutamate selected GRM5 (negative). GABA could not be regularized reliably given the small sample.

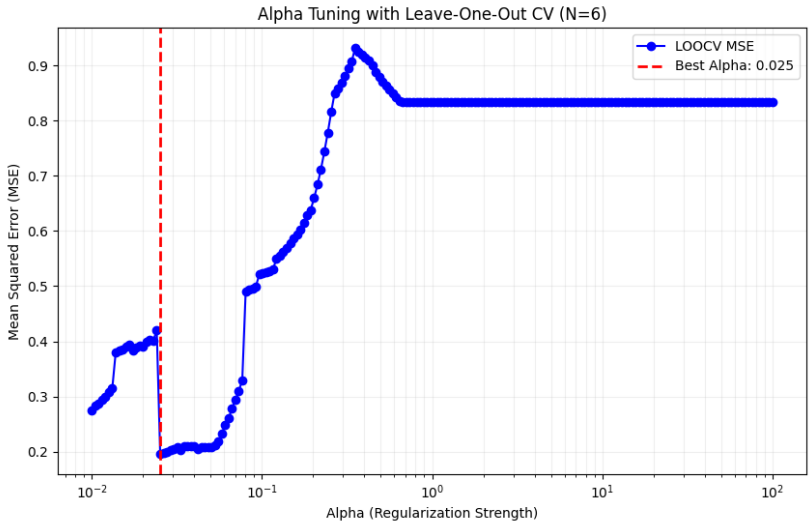


Figure 11: LOOCV MSE curve for LASSO: serotonin vs expression features.

11.2 Drug response: expression effects on score deltas

For methylphenidate recipients ($n = 5$), LASSO on score changes (post–pre) highlighted excitatory receptors (GRIA1, GRIN2B, GRM5, GRIK1) and inhibitory markers (GABRB3, GABRG3) as drivers of behavioural and hyperactivity improvements; inattention was not regularizable with the available data. Elastic-net models without drug effects on the larger untreated subset (20 subjects) selected similar glutamatergic predictors for CPRS-R domains, albeit with higher uncertainty.

11.3 Robust regressions (sensitivity checks)

HC3-robust linear models were fitted for dopamine and CPRS-R outcomes using expression and concentration blocks with sex covariates. For dopamine, no predictors were significant (all $p > 0.66$). For ADHD index vs

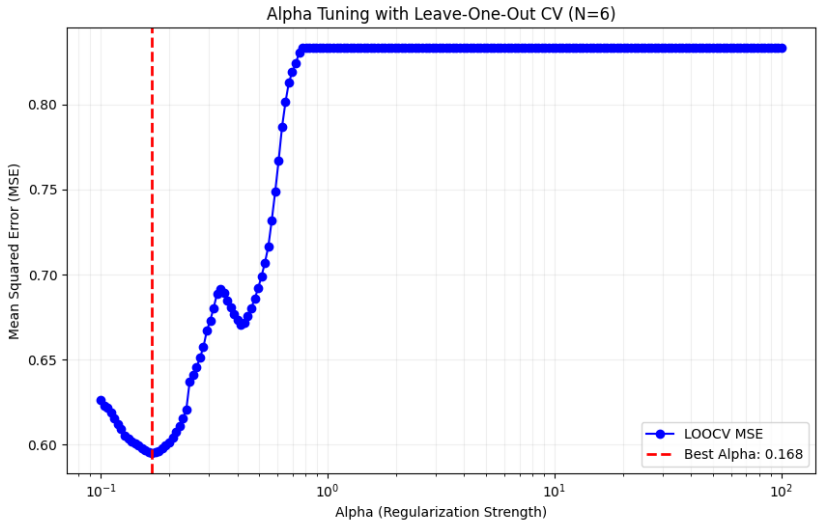


Figure 12: LOOCV MSE curve for LASSO: dopamine vs expression features.

concentrations, all transmitters were null (all $p > 0.50$). Behavioural problems showed a significant sex effect (males lower by about 10.6, $p = 0.013$) and a weak serotonin trend ($p = 0.079$). Inattention showed marginal sex effects and a small dopamine association ($p = 0.033$). These results, taken from the 'laha_out' robust regression tables, reinforce the need for sparse penalised models and highlight sex as a possible confounder.

11.4 Network structure and extensions

An exploratory Bayesian structural equation model treating expressions, concentrations, and scores as a latent network suffered from poor mixing and high prior sensitivity. A more stable extension is a factor-analytic prior aligned to glutamate, dopamine, and GABA axes, consistent with the literature-based priors in Section 8; further data are required before fitting a full joint SEM.

12 Discussion

Across modalities, cases exhibit a consistent excitation–inhibition shift: GABA receptor expression and GABA concentrations are markedly higher, glutamate concentrations are lower, and glutamatergic receptor expression (GRIN2A/B, GRM5/6/7) is elevated. MANOVA confirms that this joint shift explains most between-group variance. Penalized regressions and the logistic classifier repeatedly select the same GABAergic and glutamatergic markers, lending robustness across methods.

Imputation with numeric-only MICE minimized leakage from categorical fields while recovering usable signal; elastic-net tuning balanced sparsity and stability. Nonetheless, small sample sizes in drug-stratified analyses limit power and inflate coefficient uncertainty. HWE-filtered genotypes leave a compact but informative marker set; integrating external eQTL resources would strengthen biological interpretation.

Future work should (i) enlarge treated cohorts to stabilise drug × genotype interactions, (ii) validate pathway-level findings with Bayesian factor models that encode prior biological structure, and (iii) include longitudinal

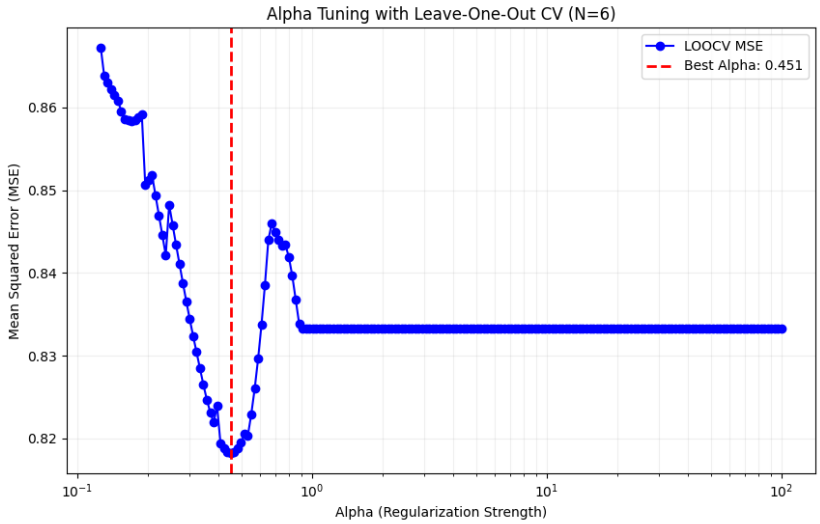


Figure 13: LOOCV MSE curve for LASSO: 5-HIAA vs expression features.

follow-up to separate treatment response from regression to the mean.

13 Conclusion & Future Work

Integrating neurotransmitter concentrations, gene expression, genotype, and clinical scores reveals a dominant GABAergic signature in neurodevelopmental cases, with complementary shifts in glutamatergic pathways. MANOVA, elastic-net regression, and logistic classification consistently identify GABA receptor genes and GABA levels as key discriminators, while glutamate shows inverse concentration and expression patterns. These findings align with prior MVK observations and support pathway-informed modelling of treatment response.

Next steps are to (i) validate SNP-specific effects via external eQTL and replication cohorts, (ii) enlarge drug-stratified samples to better characterise genotype $\times$ treatment interactions, and (iii) deploy Bayesian factor or multi-task models to encode biological priors and share information across related outcomes.

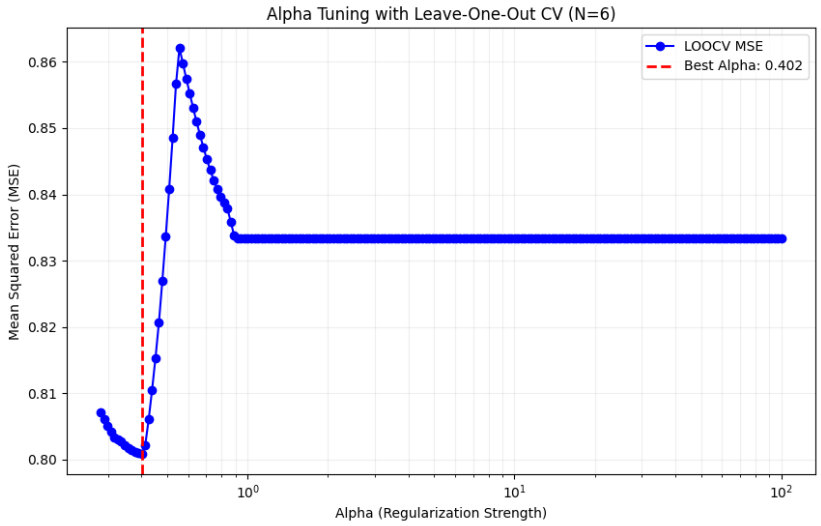


Figure 14: LOOCV MSE curve for LASSO: glutamate vs expression features.

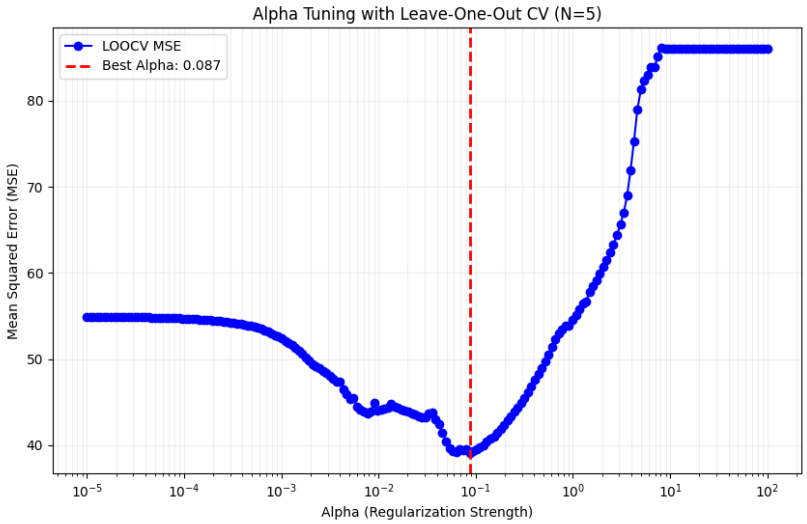


Figure 15: LOOCV for LASSO on behavioural problem change vs expressions (methylphenidate subset).

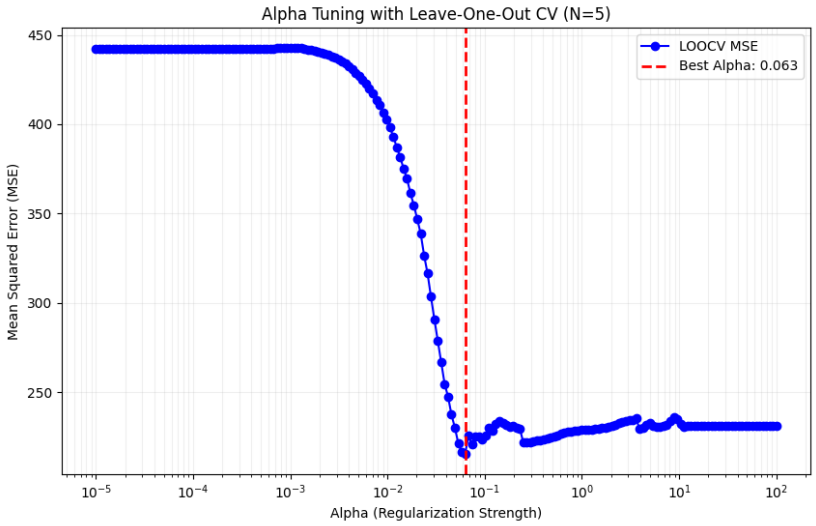


Figure 16: LOOCV for LASSO on hyperactivity change vs expressions.

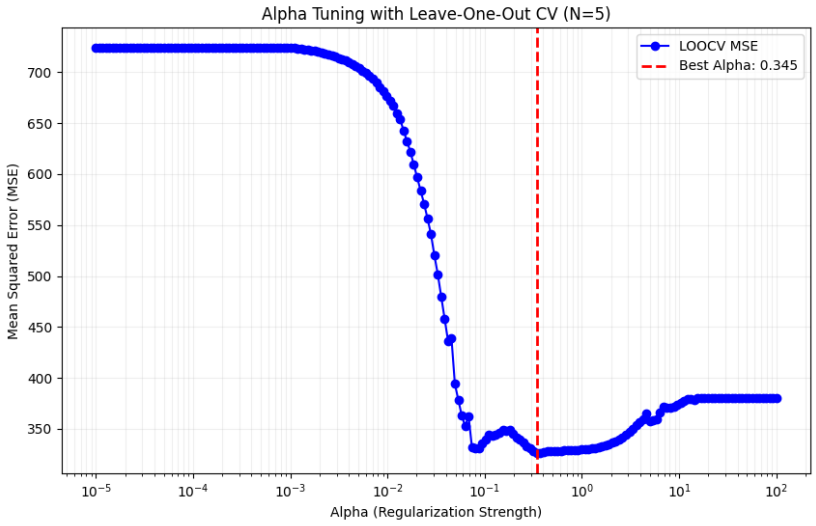


Figure 17: LOOCV for LASSO on ADHD index change vs expressions.